Q1. Describe types of wounds, complications and their management.

Q2. Discuss pathology and healing of wounds.

Q3. What are the Factors Influencing Wound Healing.

Q4. Classify wound infections. Write briefly about hospital acquired infections and their prevention

Q5. Discuss briefly about Healing by Primary Intention.

Q6. Discuss briefly about Healing by Secondary.

Q7. Diabetic Ulcer

Q8. Preauricular Sinus

Q9. What are Sinus and Fistula. What are the causes of fistula.

Q10. Lucid Interval

Q11. Keloid and hypertropic scar

Q12. Pilonidal Sinus (Jeep-bottom)

Q13. Intercostal Chest Tube (ICT) Insertion.

Q14. Abscess.

Q15.Causes of Persistent Sinus

Q1. Describe types of wounds and their management.

Types of Wounds and Their Management

Wounds encountered in surgery can be classified based on various criteria, using the mnemonic **"M-T-B-T-B"**:

1. **Morphology**
2. **Thickness**
3. **Body Involvement**
4. **Time Elapsed**
5. **Bacterial Contamination**

Classification of Wounds

1. According to Morphology

- **Open Wounds**: Skin is broken, creating a pathway to inner tissues.
- **Mnemonic: "In A Lion Pouch Gun A"** (Read as In a lion pouch gun hai)
    - *Subtypes*:
        - **Incised wounds**: Clean cuts (e.g., knife). **(Sharp bleeding)**
        - **Abrasions**: Superficial, top skin scraped (e.g., sliding fall). **(Oozing)**
        - **Lacerations**: Irregular tears from blunt force. **(Severe bleeding)**
        - **Puncture wounds**: Small holes (e.g., nail). **(Minimal bleeding)**
        - **Penetration wounds**: Object enters & exits skin (e.g., knife). **(Pathway)**
        - **Gunshot wounds**: Entry and exit wounds. **(Heavy bleeding)**
        - **Avulsions**: Tissue torn away (e.g., bite). **(Deep, heavy bleeding)**
- **Closed Wounds**: No open pathway.
- **Mnemonic: "Crush – Con – Hema" (read as Ques- Crush Con , Ans.- Hema)**
    - *Subtypes*:
        - **Crush wounds**: From prolonged heavy force.
        - **Contusions**: Bruises from blunt trauma.
        - **Hematomas**: Blood collection under the skin.

2. According to Thickness

- **Superficial wounds**: Involve *epidermis & dermis* up to papillae.
- **Partial-thickness wounds**: Involve up to *lower dermis* (hair follicles remain).
- **Full-thickness wounds**: Involve *skin & subcutaneous tissue*.
- **Deep wounds**: Include injuries to blood vessels, nerves, organs, or cavities.

3. According to Body Involvement

- **Simple wounds**: Affect one organ or tissue.
- **Combined wounds**: Affect multiple organs or tissues.

4. According to Time Elapsed

- **Fresh wounds**: Within *8 hours* of injury.
- **Old wounds**: After *8 hours* or when skin continuity is disrupted.

5. According to Bacterial Contamination

- **Clean wounds**: Made under sterile conditions (e.g., surgery).
- **Clean-contaminated wounds**: Minimal contamination from *environment or patient's skin*.
- **Contaminated wounds**: Contaminants infect the wound.
- **Dirty wounds**: Contamination from an established infection.

Chronic vs. Acute Wounds

- **Chronic wounds**: Caused by slow tissue damage (e.g., *pressure ulcers, venous ulcers*).
- **Acute wounds**: Result from trauma, disrupting tissue layers quickly.

Post-Operative Wound Complications Infection: One of the most common complications. Caused by bacteria entering the wound, leading to redness, swelling, heat, and pus formation. **Preventive measures** include proper sterile techniques during surgery and post-operative care.

- **Hemorrhage**: Bleeding from the wound site can occur immediately or delayed. It can lead to shock if significant. **Control of bleeding** during surgery and monitoring post-operatively is essential.
- **Hematoma**: Accumulation of blood under the skin that may cause swelling and pain. It can compress surrounding tissues. **Drainage** is often required for larger hematomas.
- **Seroma**: A collection of clear fluid that forms in the wound area. It can delay healing. **Aspiration** or drainage may be needed for treatment.
- **Wound Dehiscence**: Partial or total separation of wound edges, leading to exposure of underlying tissues. It may occur due to infection, poor healing, or excessive tension on the wound. **Surgical revision** is often needed.

- **Keloid and Hypertrophic Scarring**: Excessive scar tissue formation. Keloids extend beyond the wound, while hypertrophic scars stay within the wound margin. **Steroid injections or laser treatment** can be used for management.
- **Fistula Formation**: Abnormal passage between two organs or between an organ and the skin, often due to infection. **Surgical intervention** is required for treatment.
- **Delayed Healing**: Caused by factors like poor nutrition, infection, or poor blood supply. **Proper wound care** and managing underlying conditions can help promote healing.
- **Evisceration**: Protrusion of internal organs through the wound. It is an emergency situation requiring immediate **surgical intervention**.
- **Pain**: Post-operative pain is common, but excessive pain may indicate complications such as infection or hematoma. **Pain management** should be adjusted based on the severity.

Key Points to Prevent Complications

- Good **sterile technique** during surgery
- Careful **wound monitoring** post-surgery
- Addressing underlying health conditions like diabetes or poor nutrition
- Providing **proper wound care** and hygiene

Management of Wounds **(Any type of wound mentioned in question)**

Wound management varies based on the **type**, **cause**, **depth**, involvement of other structures, and **time since injury**.

A) Open Wound Management

1. **General Steps**: **Mnemonic:** **"PCD – Se – M**aine (Read as PCD Se Maine Open wound management kiya)
    - **Pressure**: Apply pressure to stop bleeding.
    - **Cleaning**: Use sterile saline, antiseptic solution, or hydrogen peroxide.
    - **Dressing**: Cover wound with a sterile dressing.
    - **Stitching**: Fresh wounds (within 12 hours) can be **sutured** or stapled.
    - **Medication**: Antibiotics to prevent infection.
2. **Specific Types**:
    - **Lacerations**: Examine, clean, trim edges, and close the wound. *Wounds older than 24 hours* are likely contaminated, so only deep layers are closed, and **skin is left open**.
    - **Abrasions**: Usually heal on their own with skin discoloration fading in **1-2 weeks**. Keep the area **clean** with soap and water; no active treatment needed.
    - **Puncture Wounds**: Entry point is **left open** to allow drainage of bacteria or debris, as these wounds are prone to infection.
    - **Bite Wounds**: Avoid suturing unless a large area or face is involved, as bites often get infected. **Leave open** for daily wound care. *Antibiotics are necessary*.

B) Closed Wound Management

1. **Contusions (Bruises)**:
    - **Rest**: Limit movement of injured site.
    - **Ice/Cold Compress**: Apply for 10-15 minutes every 2 hours to reduce blood flow.
    - **Compression**: Wrap with a bandage to control swelling.
    - **Elevation**: Raise above heart level to reduce blood pooling.
2. **Hematomas**:
    - Treatment similar to contusions.
    - **Elastic Bandage**: Apply pressure to reduce swelling.
    - **Antibiotics**: Recommended to prevent infection.
3. **Crush Wounds**:
    - **Do Not Move**: Avoid moving the injured person.
    - **Treat for Shock**: Keep warm if shock is present.
    - **Compression**: Start compression on the injured area.
    - **Medical Interventions**: May involve IV fluids, hyperbaric oxygen therapy, and, if severe, **amputation**.

Q2. Discuss pathology and healing of wounds.

Pathology and Healing of Wounds

Wound healing involves a series of events that happen in both **acute** and **chronic wounds**. These stages include:

1. **Inflammation**
2. **Granulation tissue formation & revascularization**
3. **Epithelialization**
4. **Wound contraction**
5. **Scar formation**

Phases of Wound Healing

1. **Inflammatory Phase**
    - **Duration**: Begins at injury and lasts 2-4 days.
    - **Key Events**: Haemostasis (stopping bleeding) and inflammation.
    - **Process**:
        - **Platelet Plug Formation**: Platelets form a plug at the wound site.
        - **Vasoconstriction & Clotting**: Calcium and tissue factor activate clotting; **fibrin plug** forms.
        - **Vasodilation**: Histamine release allows immune cells to reach the wound.
        - **Key Cells**:

        - **Platelets**: Release PDGF & TGF-B to attract neutrophils and macrophages.
        - **Neutrophils**: Clean bacteria and debris.
        - **Macrophages**: Continue cleaning and release growth factors to attract **fibroblasts** for the next phase.
2. **Proliferative Phase**
    - **Duration**: Begins around day 3, overlaps with the inflammatory phase.
    - **Key Events**: Angiogenesis, collagen deposition, granulation tissue formation, epithelialization.
    - **Process**:
        - **Fibroblasts**: Initiate formation of new blood vessels, collagen, and epithelial tissue.
        - **Granulation Tissue Formation**: Fibroblasts produce **collagen**, glycosaminoglycans, and proteoglycans to form granulation tissue.
        - **Epithelialization**: New skin cells spread over the wound from the edges or basement membrane.
        - **Type of Collagen**: Mainly **Type III collagen** produced in this phase.
3. **Remodeling Phase**
    - **Duration**: Lasts 6-12 months.
    - **Key Events**: Collagen reorganization, strengthening of tissue.
    - **Process**:
        - **Collagen Remodeling**: Type III collagen is replaced with **Type I collagen**.
        - **Tensile Strength**: Tissue reaches about 80% of original strength after a year.
        - **Myofibroblasts**: Cause wound contraction.
        - **Decreased Vascularity**: Wound becomes less red and more aesthetically pleasing.

Types of Wound Healing

1. **Primary Intention (Primary Healing)**
    - **Process**: Skin edges are surgically closed (e.g., sutures, skin grafts).
    - **Use**: Small, clean wounds where edges can be brought together.
2. **Secondary Intention (Secondary Healing)**
    - **Process**: Wound is left open to heal by forming granulation tissue and re-epithelialization.
    - **Use**: Wounds like abrasions or where skin can't be closed.
3. **Tertiary Intention (Delayed Primary Closure)**
    - **Process**: Wound is first cleaned and left open, then later closed surgically or with grafts.
    - **Use**: Contaminated wounds needing infection control before closure.

Q3. What are the Factors Influencing Wound Healing.

Local Factors:

"Big Doctors In Medical Practice Take Surgical Care"

1. **Blood Supply/Oxygen Supply**
   - Adequate blood and oxygen are crucial for healing.
2. **Denervation**
   - Decreased wound contraction if nerves are damaged.
3. **Infection**
   - Prolongs inflammation and delays healing.
4. **Mechanical Stress**
   - Affects collagen fibers' quantity and orientation.
5. **Pressure**
   - Undue pressure can cause ischemia, delaying healing.
6. **Types of Tissue**
   - Different tissues have different healing abilities.
7. **Surgical Technique and Material**
   - Correct technique and materials (like sutures) aid healing.

General Factors:

"A NICE CLOTH haveVery Tiny Foreign bodies from Regular Smoking"

- **Age**
  - Older age slows healing and reduces tensile strength.
- **NSAIDs**
  - Decrease collagen synthesis.
- **Infection**
  - Pus lowers pO2, inhibits epithelialization, and prolongs inflammation.
- **Cytotoxic Drugs**
  - Affect WBCs, fibroblast proliferation, and collagen synthesis.
- **Environmental factors (Temperature)**
  - Higher temperatures >30°C improve tensile strength.
- **Corticosteroids**
  - Inhibit macrophages, inflammatory response, and collagen formation.
- **Low blood volume (Hypovolemia)**
  - Reduced blood volume leads to poor perfusion and healing.
- **Obesity**
  - Impairs wound healing and increases infection risk.
- **Trace Metal Deficiency**
  - **Zinc** deficiency impairs collagen synthesis and immune function.
- **Hypoxia**
  - Low oxygen levels impair collagen formation and immune response.
- **Vitamin Deficiency**

    - Lack of vitamins like A, C, and E affect collagen and healing.
- **Temperature**
- **Foreign Body**
    - Clot, dirt, or sutures can cause infection and slow healing.
- **Radiation Therapy**
    - Acute radiation damages blood vessels; chronic radiation causes irreversible skin damage.
- **Smoking**
    - Delays healing by reducing oxygen supply to tissues.

Q4. Classify wound infections. Write briefly about hospital acquired infections and their prevention.

Classification of Wound Infections

Wound infection: Bacterial multiplication in tissue with a **host reaction**.

1. **Superficial Wound Infections**:
    - Involves only **skin and subcutaneous tissue**.
    - Signs of **inflammation**.
2. **Deep Wound Infections**:
    - Defined at **30 days post-surgery** or **1 year if implant involved**.
    - Infection present in **deep soft tissues**.
3. **Organ/Space SSIs (Surgical Site Infections)**:
    - Affects parts of the **anatomy beyond the incision**.

Major Pathogens in Wound Infections:

- **Staphylococcus aureus**, coagulase-negative staphylococci
- **Enterococci**
- **Escherichia coli**
- **Pseudomonas aeruginosa**
- **Enterobacter** species
- **Proteus mirabilis**
- **Klebsiella pneumoniae**

Wound Contamination:

1. **Direct Contact**: From **equipment or hands** of carriers.
2. **Airborne Dispersal**: From **surrounding air**.
3. **Self-Contamination**: From **patient's own skin or gastrointestinal tract**.

Clinical Signs of Wound Infection:

- **Localized erythema** (redness)
- **Localized pain and heat**
- **Cellulitis**
- **Oedema**
- **Abscess** formation
- **Discharge**: Viscous, **discolored, purulent**
- **Delayed healing**
- **Tissue discoloration** at wound margins
- **Friable, bleeding granulation tissue**
- **Unexpected pain/tenderness**
- **Abnormal smell**
- **Uneven granulation tissue**

Treatment:

- **Antibiotics**
- **Iodine, silver application**
- **Autolytic/enzymatic debridement**
- **Surgical debridement**
- **Maggot therapy**

Hospital-Acquired Infections (Nosocomial Infections)

Definition: Infections from hospital/healthcare settings, appearing **48+ hours post-admission** or **within 30 days after discharge**.

Common Types:

1. **Surgical wound and soft tissue infections**
2. **Urinary tract infections**
3. **Respiratory infections**
4. **Gastroenteritis**
5. **Meningitis**

Pathogen Types:

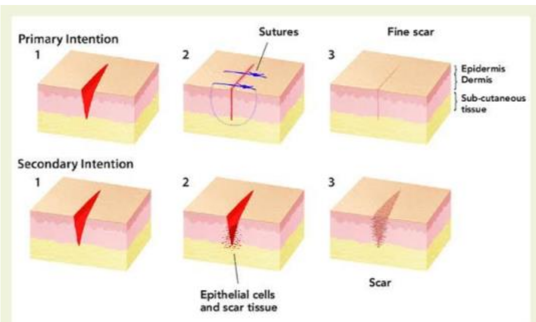
- **Conventional Pathogens**: Cause disease in **healthy individuals**.
- **Conditional Pathogens**: Cause disease in **lowered resistance** cases.
- **Opportunistic Pathogens**: Cause disease in **severely immunocompromised** patients.

Prevention of Nosocomial Infections:

- **Isolation** of infected patients
- **Handwashing and gloving**

- **Surface sanitation**: Methods like **NAV-CO2**, **hydrogen peroxide vapour** for effective cleaning.
- **Use of aprons**
- **Safe injection practices**
- **Environmental cleaning**
- **Use of hot/superheated water**
- **Disinfection of patient equipment**
- **Sterilization**

Q5. Discuss briefly about Healing by Primary Intention.



Healing by primary intention: **Fibrous adhesion** occurs without **suppuration** (pus formation) or **granulation tissue** development.

Characteristics of Primary Intention Healing:

- **Uninfected and clean wound**
- **Surgical incision**
- **Minimal tissue loss**
- **Wound edges** are brought together with **sutures**

Sequence of Events in Primary Intention Healing:

1. **Bleeding/Hemorrhage**:
    - Blood fills the space between the wound edges, held by **sutures**.
    - Blood vessels are disrupted, exposing **collagen**, which activates **platelets**.
    - **Clot formation** occurs to seal the wound, preventing **dehydration and infection**.
2. **Acute Inflammation**:
    - Begins within **24 hours** and marked by **polymorphs** (immune cells).
    - **Vasodilation** (widening of blood vessels) replaces initial **vasoconstriction**.
    - **Capillary permeability** increases, allowing **plasma and leucocytes** to enter and clear bacteria/debris.
    - **Macrophages** release growth factors to attract **fibroblasts** for the next healing phase.

3. **Proliferation**:
   - **Basal cells** from both wound edges proliferate and migrate to close the wound space.
4. **Remodeling**:
   - **Matrix breakdown** and **new matrix formation** occur continuously.
   - **Fibroblasts** invade the area by the **3rd day**, producing **collagen fibers** by the **5th day**.
   - By the **4th week**, **scar tissue** forms, with minimal cells, vessels, and some remaining inflammatory cells.

Contraindications for Healing by Primary Intention:

- **Acute wounds >6 hours old** (except facial wounds)
- **Foreign debris** that can't be fully removed
- **Active bleeding**
- **Dead space** under the skin closure
- **Excessive tension** on the wound

Q6. Discuss briefly about Healing by Secondary intention.

Healing by secondary intention: **Union of two wound surfaces** through **granulation** and **suppuration** (pus formation).

Characteristics of Secondary Intention Healing:

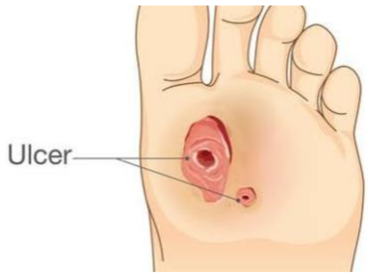
- **Extensive tissue and cell loss**
- **Open, large tissue defect**
- **Dehisced (reopened) surgical wounds**
- Wounds from **underlying conditions** (e.g., pressure necrosis, chronic venous insufficiency)

Sequence of Events in Secondary Intention Healing:

1. **Minimal or No Hemorrhage**:
   - **Bleeding and clot formation** are absent or minimal, which slows the healing process.
   - **Surgical debridement** may be used to initiate bleeding, aiding in the **repair process**.
2. **Prolonged Inflammation**:
   - Extended due to **necrotic tissues** and **higher bacterial load**.
   - Inflammation lasts longer than the **3 days** seen in primary healing, continuing until **necrotic tissue** is cleared and infection is under control.
3. **Prolonged Proliferation**:
   - Differs from primary healing:
     - **Granulation tissue formation** fills the tissue defect.

        - **Contraction** reduces the size of the defect.
        - **Epithelialization** is delayed and occurs last.
4. **Prolonged Remodeling**:
    - Similar to primary intention but **more extended**.
    - **Tensile strength** of the wound increases over **2-3 months**.

Q7. Diabetic Ulcer



Diabetic ulcers: Chronic **foot ulcers** common in diabetics.

Pathogenesis:

- Caused by **neuropathic impairment** (nerve damage) and **immune compromise**.
- **Peripheral vascular disease** further complicates these ulcers with **infection**.
- **Pressure and tissue trauma** are contributing factors, but **neuropathy** from diabetes is key.
- Causes of diabetic ulcers:
    - **Neuropathy**: 60-70%
    - **Ischaemia** (restricted blood flow): 15-20%
    - **Combination of both**: 15-20%
- **Neuropathy** involves:
    - **Sensory loss**: Leads to unrecognized injuries (e.g., from shoes or foreign bodies), increasing stress on areas like **metatarsal heads**, **heels**, and **callosities**.
    - **Motor neuropathy (Charcot foot)**: Causes joint collapse/dislocation, adding pressure on unprotected areas and severe **vascular impairment**.

Treatment:

- **Local and systemic measures** are essential.
- **Standard care**:
    - **Off-loading** (pressure relief)
    - **Debridement** (removal of dead tissue)
    - **Moist wound environment**
    - **Systemic antibiotics** for cellulitis.
- **Blood sugar control** is critical.

- **Infection management**:
  - Most ulcers are infected; clearing the infection is crucial.
  - **Antibiotics** targeting both **soft tissue** and **bone** to address possible **osteomyelitis**.
  - **Wide debridement** of all necrotic or infected tissue.
- **Off-loading** using specialized orthotic shoes or casts to protect the ulcer.
- **Topical treatments** like **PDGF** and **granulocyte-macrophage colony-stimulating factor** can aid in closure.

Prevention:

- **Foot care** and **prevention strategies** are key in managing diabetic ulcers.

Q8. Preauricular Sinus

A **preauricular sinus** is a **benign congenital malformation** of the soft tissues near the ear. It may also be called a **preauricular pit**, **fistula**, **tract**, or **cyst**.

Inheritance:

- Typically follows an **incomplete autosomal dominant** pattern.
- May occur **spontaneously**.
- Can be **bilateral or unilateral**.

Pathophysiology:

- Results from **incomplete fusion** of 2 of the **6 hillocks** formed by the **first and second branchial arches**.

Clinical Features:

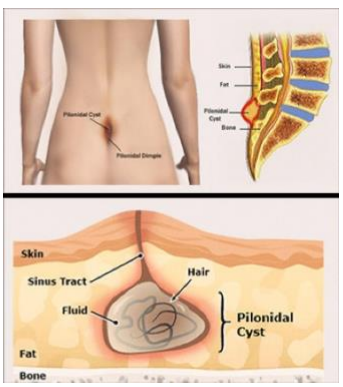
- Often **asymptomatic**.
- Appears as a **small pit** near the external ear, usually at the **anterior margin of the ascending helix**.
- **Pit location** can also be seen at the **posterosuperior helix margin**, **tragus**, or **lobule**.

- The pit may lead to a **sinus tract**, which can vary in **length and direction**.
- Some patients may experience **chronic drainage** of pus, and once infected, the sinus may become **symptomatic** with **recurrent infections**.
- Infection can lead to **facial cellulitis** or **ulcerations** near the ear.

Treatment:

- Once infection occurs, **recurrent exacerbations** are common.
- **Surgical removal** of the sinus tract is recommended after treating infection with **antibiotics** and allowing inflammation to subside.
- Standard removal involves an **incision around the sinus** and careful dissection to remove the entire tract.
- The **supra-auricular approach** may offer higher success for complete removal.

Q9. What are Sinus and Fistula. What are the causes of fistula.



Sinus:

- A **sinus** refers to a **chronically infected tract** that connects an **abscess** to the skin. It is often formed due to infection or inflammation, creating a tunnel-like passage.
- **Sinus tracts** often result from abscesses or ongoing infections, typically in the skin or internal organs.
- Common example: **Pilonidal sinus**, a condition involving an infected sac at the base of the spine.

Fistula:

- A **fistula** is an **abnormal connection** or passageway that forms between **two epithelium-lined organs** or vessels that do not normally connect.
- Fistulas are usually **pathological**, resulting from disease, infection, or injury. However, they can also be **surgically created** for therapeutic purposes.

- Types of **fistulas**:
    1. **Blind Fistula**: Has **only one open end** and is not connected to any internal organ or vessel.
    2. **Complete Fistula**: Has **both external and internal openings**, connecting two organs or parts of the body.
    3. **Incomplete Fistula**: Has an **external opening** (usually on the skin) but does not connect to any internal organ or structure.

Causes of Fistulas: Fistulas can develop due to various **disease conditions**, **medical treatments**, or **trauma**. Some common causes include:

1. **Diseases**:
    - **Inflammatory bowel diseases** like **Crohn's disease** and **ulcerative colitis** can cause **anorectal fistulas** (fistulas between the anus and surrounding skin).
    - **Infections**: Conditions like **tuberculosis** or **actinomycosis** may lead to fistulas forming due to chronic infection.
2. **Trauma/Injury**:
    - **Prolonged or difficult childbirth** can lead to **obstetric fistulas** (abnormal connections between the vagina, bladder, or rectum, often due to pressure during labor).
    - **Surgical complications**: Incorrectly performed surgeries can inadvertently create fistulas between organs.
    - **Accidental injuries**: Trauma from accidents, knife wounds, or gunshot wounds can cause abnormal connections between tissues or organs.
3. **Medical Treatments**:
    - Certain **medical procedures** (e.g., organ transplants, cancer treatments) may create fistulas intentionally, like in the creation of **dialysis fistulas** for hemodialysis patients.
4. **Chronic Infections**:
    - Chronic **abscesses** or infections can form a **sinus** that may evolve into a fistula, especially when left untreated.

Key Points:

- **Sinus**: Infected tract linking an abscess to the skin.
- **Fistula**: Abnormal passageway between two normally unconnected organs or vessels.
- Types of fistulas: **Blind**, **Complete**, **Incomplete**.
- **Common causes**: **Crohn's disease**, **trauma**, **difficult childbirth**, **surgical errors**, and **chronic infections**.

Q10. Lucid Interval

- **Definition**:
  A **lucid interval** is a significant clinical sign in neurology and is often considered a hallmark of **intracranial hematomas**, especially **epidural hematoma**. It is characterized by a period of regained consciousness after an initial loss of consciousness due to brain injury.

- **Mechanism**:
  1. **Initial impact**:
     - A head injury causes **temporary unconsciousness** due to trauma to the brain.
  2. **Lucid phase**:
     - The patient regains **full consciousness**, appears normal, and may not show any severe symptoms initially.
     - This phase is due to the delayed effects of bleeding or pressure buildup.
  3. **Deterioration**:
     - As bleeding or swelling progresses, **intracranial pressure (ICP)** rises, leading to a rapid decline in the patient's condition.

- **Causes of Lucid Interval**:
  1. **Epidural hematoma**:
     - Most commonly associated with a **tear in the middle meningeal artery** following trauma.
  2. **Subdural hematoma**:
     - Less commonly presents with a lucid interval but may occur.
  3. **Brain edema**:
     - Swelling in the brain that temporarily resolves before worsening.
  4. **Multiple contusions**:
     - Bruising in different parts of the brain leading to delayed symptoms.

- **Clinical Significance**:
  - The lucid interval is a **critical warning sign** that requires **immediate medical attention**.
  - If untreated, the patient's condition can worsen, leading to:
    - **Severe neurological deficits**
    - **Coma**
    - **Death** due to brain herniation.

- **Symptoms in Lucid Interval**:
  - Initially **unconsciousness** due to the injury.
  - Recovery to a **normal or near-normal state** (lucid period).
  - Later, progressive symptoms such as:
    - **Severe headache**
    - **Vomiting**

        - **Drowsiness or confusion**
        - **Hemiparesis (weakness on one side of the body)**
        - **Seizures**

- **Key Takeaway**:
    - The **lucid interval** is a deceptive phase that can delay treatment. Any head injury patient showing such signs must be promptly evaluated with imaging like **CT or MRI** for early diagnosis and management.

Q11. Keloid and hypertropic scar.

- **Keloids** are **excessive growth** of **scar tissue** that extends beyond the original wound area.
- **Benign** (not cancerous), **non-contagious**, and do not shrink or resolve over time.
- **Formed** due to **overgrowth of granulation tissue** (mainly **collagen type III**) at the site of a healed skin injury.
- This granulation tissue is later replaced by **collagen type I**, leading to the raised scar.

Characteristics of Keloids

- **Symptoms**:
    - Can cause **severe itching** and **pain**.
    - Changes the **texture** of the skin.
    - **Shiny and raised** appearance.
- **Common in certain races**:
    - More frequent in **Afro-Caribbean** and **Oriental** races.

Common Sites

- **Chest** (center).
- **Back**.
- **Shoulders**.
- **Ear lobes**.

Treatment

- **Pressure application**: To help reduce size and flatten keloid.
- **Steroid injections**: **Triamcinolone** is injected into the keloid to reduce inflammation and size.
- **Surgery**: Removal of the keloid, though it may recur.

Hypertrophic Scar

- **Definition**: A **raised, red (erythematous), itchy (pruritic)** lesion that stays **within the boundaries** of the original scar.
- **Histology**: Contains **collagen fibers** arranged in **nodules** with **myofibroblasts** and an **increased number of blood vessels**.
- **Features**:
  - Looks similar to a **keloid** but **does not spread** into surrounding tissues.
  - **Regresses naturally** over time.
  - Commonly found **over joints** due to movement and tension.
- **Causes**:
  - **Injury to the skin** (cuts, burns, surgery, or trauma).
  - **Inflammatory skin conditions** like acne.
  - **Genetic predisposition** in certain individuals.

Differences Between Keloid and Hypertrophic Scar

Feature	Hypertrophic Scar	Keloid
Spread	Confined to original scar	Extends beyond original scar
Regression	Regresses spontaneously	Does not regress
Appearance	Less raised, erythematous	More raised, dark in color
Common Sites	Joints	Ear lobes, chest, shoulders

Management

- **Non-surgical**:
  - **Pressure therapy**: Use of compression garments.
  - **Silicone gel sheets**: Helps reduce redness and flatten the scar.
  - **Corticosteroid injections**: Reduce inflammation and scar size.
  - **Laser therapy**: Improves appearance and reduces vascularity.
- **Surgical**:

    - **Excision**: Removal of the scar followed by preventive treatments to avoid recurrence.

Prevention

- **Wound care**: Proper cleaning and dressing of wounds.
- **Avoidance of tension** on sutures or wounds during healing.

Use of **silicone gels** or sheets during early stages of scar formation

Q12. Pilonidal Sinus (Jeep-bottom)

- **Meaning**: "Nest of hairs" (from Greek).
- Called **Jeep-bottom** as it was common in jeep drivers.
- More common in **hairy men** and **dark-skinned** individuals.
- Appears usually between ages **20-30 years**.
- **Acquired condition** caused by **vibration** and **friction**, which makes hair collect in the **gluteal cleft** (between buttocks) and enter sweat gland openings.
- Hair forms a **blind-ended sinus** (one closed end).

Clinical Features

- **Location**: Sinus opening just above the **anal verge**, in the midline over the **coccyx** (tailbone area).
- **Symptoms**:
    - **Pus discharge**.
    - History of **recurrent abscesses** (pockets of pus) that rupture.
    - **May be asymptomatic** (no symptoms).

Diagnosis

- **Differential diagnosis**: Only condition to consider is **osteomyelitis of the coccyx**.
- **X-ray of the coccyx** is recommended to confirm diagnosis.

Treatment

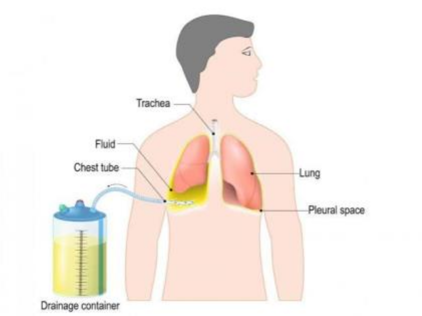
1. **Injection of methylene blue**: Used to mark sinus branches before excision.
    - Patient positioned **prone with buttocks elevated** (**Jack-knife position**).
2. **Methods for wound treatment**:
    - **Open method**:
        - **Leave wound open** after excision.
        - Pack with **iodine or eusol gauze**.
        - Takes **3-4 weeks to heal**.

        - **Sitz bath** (warm water bath) recommended.
        - **Least recurrence**.
    - **Closed method**:
        - Close wound using **Z-plasty** or **Rhomboid (Limberg) flap**.
        - **10-20% recurrence rate**.
    - **Karydakis procedure**:
        - Removes all sinus tracts till **sacral bone**.
        - **Semilateral incision** made around sinus.
        - **Tension-free closure** reduces skin necrosis risk.
    - **Bascom's technique**:
        - **Lateral incision**, avoids midline.
        - Excision of infected sinuses.
        - **Lateral wound left open** (instead of midline).

Key Points

- **Multiple sinuses** may communicate with each other.
- Sinuses usually have a **cephaloid direction** (upwards).
- **Recurrent abscesses** are common.
- **Treatment options** include excision with or without **marsupialization**, **flap closure**, or **Z-plasty**.
- **Recurrence is common** even with good surgical methods.

Q13. Intercostal Chest Tube (ICT) Insertion.



Purpose: To drain **air (pneumothorax)** or **fluid (haemothorax/empyema)** from the pleural cavity.

Ideal Sites

1. **2nd Intercostal Space (ICS)** at the **midclavicular line**: For **pneumothorax**.
2. **6th ICS** at the **midaxillary line**: For **haemothorax**.

Triangle of Safety

- **Above** the level of **nipples**.
- **Anterior** to the **midaxillary line**.
- **Below and lateral** to the **pectoralis major muscle**.

Procedure

1. **Position the patient**: Supine, with head elevated.
2. **Infiltrate local anaesthetic**: Up to the **parietal pleura**.
3. **Incision**:
    - 2-3 cm **parallel to ribs**.
    - Use **sharp dissection** to cut the skin.
4. **Blunt dissection**:
    - Separate **serratus** and **intercostal muscles** using finger and artery forceps.
    - Reach the **pleura**.
5. **Insert the chest tube**:
    - With **trocar** into the pleural cavity.
    - Remove trocar, connect tube to **underwater seal**.
6. **Fix the tube**:
    - Use a **suture** to secure.
    - Create a short **oblique tunnel** to prevent leaks.
7. **Monitor**: Observe fluid column movement with respiration.
8. **Chest X-ray**: Confirm proper placement and lung expansion.

Ten Commandments for ICT

1. Use the **triangle of safety**.
2. For:
    - **Pneumothorax/haemothorax**: Direct tube to **apex**.
    - **Empyema**: Direct tube to **base**.
3. Ensure **all holes** in the tube are inside the pleural cavity.
4. Connect to an **underwater seal**.
5. Check **fluid column movement** with breathing.
6. Confirm placement with a **chest X-ray**.
7. Avoid **kinking** of the tube.
8. Do **not clamp** the tube if there's an air leak.
9. Keep the drainage system **below patient level** to avoid backflow.
10. Never allow bottle contents to re-enter pleural space.

Removal of ICT

- **Prerequisites**:

- **Chest X-ray**: Lung fully expanded.
- Drainage < **100 ml/day for 2 days**.
- No air leak.
- Patient off ventilator.

- **Steps**:

1. **Clamp the tube** for **24 hours**.
2. Remove tube if:
   - Patient remains comfortable.
   - Lung stays expanded.

Q14. Abscess.

- **Definition**: A localized collection of **pus** (dead and dying neutrophils + proteinaceous exudate).

Classification

1. **Pyogenic Abscess**:
   - Most common type.
   - Can be **subcutaneous, deep**, or in viscera (e.g., liver, kidney).
2. **Pyaemic Abscess**:
   - Caused by circulation of **pyaemic emboli** in blood (**pyaemia**).
3. **Cold Abscess**:
   - Seen in **tuberculosis**, involving lymph nodes or spine.

Pyogenic Abscess

- **Caused by**:
  - **Staphylococcal infection** via wounds or hematogenous spread (e.g., tonsillitis, carious teeth).
  - Can also result from **cellulitis**.

Pathophysiology

1. **Injury** → **Inflammation** caused by organisms (e.g., Staphylococcus).

2. Formation of **pus**:
    - Contains **dead leukocytes, bacteria**, and necrotic tissue.
3. **Pyogenic membrane**:
    - Fibrin products encircle the abscess, infiltrated by leukocytes and bacteria.

Symptoms

- **Throbbing pain**: Indicates pus; caused by pressure on nerve endings.
- **Fever**: With or without chills and rigors.
- **General illness**.

Signs (5 Classical Signs)

1. **Calor (Heat)**: Local rise in temperature.
2. **Rubor (Redness)**: Inflammation causes **hyperaemia**.
3. **Dolor (Pain)**: Extremely tender abscess.
4. **Tumor (Swelling)**:
    - Tense and cystic.
    - Surrounded by **brawny oedema**.
5. **Loss of Function**: Impaired due to pain.

- **Fluctuation**: Positive in superficial abscess; negative in deep ones (e.g., breast abscess).

Treatment

1. **Natural course**:
    - Points along the path of **least resistance** to reach epithelial surface (e.g., skin, gut).
    - **Deep abscesses** (e.g., breast) may cause significant tissue destruction before pointing.
2. **Incision and Drainage (I&D)**:
    - Done under **general anaesthesia** (preferred).
    - **Why GA?**:
        - Local anaesthesia is ineffective in infected tissues.
        - Breaking loculi without causing pain is challenging.

Procedure for Incision & Drainage (I&D)

1. **Stab Incision**:
    - Made over the **most prominent (pointing)** part of the abscess.
    - Collect pus for **culture and sensitivity**.
2. **Breaking Loculi**:

    - Use **sinus forceps or finger** to break all loculi.
    - Fresh oozing of pus indicates completion.
3. **Irrigation**:
    - Irrigate cavity with **saline** or mild antiseptic agents (e.g., iodine solution, hydrogen peroxide).
    - **H_2O_2 reaction**:
        - Liberates **nascent oxygen** → separates slough.
            - $H_2O_2 \rightarrow H_2O + [O]$.
4. **Packing**:
    - Large cavities are packed with **roller gauze dipped in iodine solution**.
    - Packing prevents **premature closure of skin**, ensuring healing by granulation tissue.
5. **Healing Time**:
    - With **antibiotics** and proper dressing, healing occurs in **5-7 days**.
6. **Antibiotic of Choice**:
    - **Cloxacillin** for staphylococcal abscess.
    - Dosage: **500 mg, 6th hourly for 5-7 days**.
7. **Modified Hilton's Method**:
    - Used near vital structures (e.g., **vessels, nerves**).
    - **Procedure**:
        - Incise **skin and superficial fascia**.
        - Use sinus forceps to open abscess → prevents damage to vessels/nerves.

Differential Diagnosis of Abscess

1. **Ruptured Aneurysm**:
    - May mimic an abscess with pain, redness, and leukocytosis.
    - Common sites:
        - **Posterior triangle (vertebral artery)**.
        - **Popliteal fossa (popliteal artery)**.
    - **Caution**: Aspirate with a wide-bore needle if in doubt.
2. **Soft Tissue Sarcoma**:
    - Can mimic deep-seated abscess.
    - **Key differences**: No throbbing pain or high-grade fever.

Antibioma

- **Definition**: Antibiotic-induced swelling.
- **Cause**: Partial sterilization of pus + wall fibrosis due to antibiotics.
- **Sites**: Breast, thigh, ischiorectal fossa.
- **Breast antibioma** can mimic **carcinoma**.

Pyaemic Abscess

Definition:

A collection of pus caused by the circulation of pyogenic (pus-forming) organisms in the bloodstream, leading to the formation of multiple abscesses.

Causes:

•Circulation of pus-producing organisms in the blood (pyaemia).

•Common conditions predisposing to pyaemic abscess:

- **Diabetics**: Reduced immunity increases susceptibility to infections.
- **Patients on chemotherapy or radiotherapy:** Immunosuppression leads to easy spread of infections.

Clinical Features:

- **Multiple abscesses:** Often occur in various parts of the body due to hematogenous spread.
- **Deep-seated lesions:** Located in deep tissues or organs.
- **Minimal tenderness**: Unlike typical abscesses, tenderness is usually mild or absent.
- **Nonreactive abscess (No local rise in temperature)**: Since the immune response is blunted, classical inflammatory signs like redness and warmth are not prominent.

Diagnosis:

- **Imaging**: CT scan or ultrasound to detect deep-seated abscesses.
- **Blood cultures**: To identify the causative organism.
- **Complete Blood Count (CBC):** May show leukocytosis.

Treatment:

- **Multiple incisions and drainage:** To evacuate pus and reduce pressure.
- **Antibiotics**: Broad-spectrum antibiotics targeting pyogenic organisms.
- **Management of underlying condition**: Control of diabetes or discontinuation of immunosuppressive therapies if possible.

Cold Abscess

Definition:

A chronic abscess characterized by the **absence of classical signs of inflammation** (redness, heat, pain, and swelling). Primarily associated with **granulomatous infections.**

Causes:

- **Most common cause:** Tuberculosis
- **Other causes:**
    - **Leprosy:** Chronic bacterial infection affecting skin and nerves.
    - **Actinomycosis**: Rare bacterial infection causing abscesses and draining sinuses.
    - **Madura foot:** Chronic fungal infection of the foot, causing abscesses and disfigurement.

Tuberculosis as the Leading Cause:

- **Tubercular lymphadenitis:** Involves **cervical lymph nodes**. Cold abscess often forms in the **neck region.**
- **Pott's disease:** Tuberculosis of the spine. Cold abscess forms in the **paraspinal region** and may track to distant areas like the **psoas muscle.**

Clinical Features:

- **Chronic swelling: Fluctuant and painless.**
- **Discharging sinus**: May rupture spontaneously, discharging **caseous material.**
- **Minimal systemic signs:** Fever and malaise are usually absent or minimal.

Diagnosis*:*

- **Imaging:**
    - **X-ray/MRI:** Detect bony lesions (e.g., Pott's disease).
    - **Ultrasound**: For soft tissue abscesses.
- **Aspiration cytology/biopsy:** Detects **granulomas** or **acid-fast bacilli (AFB)** in tuberculosis.
- **Tuberculin Skin Test:** Supports TB diagnosis.

Treatment:

- **Antitubercular therapy (ATT): 6-9 months** of drugs like **isoniazid, rifampicin, pyrazinamide, and ethambutol.**
- **Drainage:** For large abscesses causing compression or discomfort.
- **Surgical excision: For resistant or complicated cases.**

Differential Diagnosis:

- **Lipoma:** If no discharge is present.
- **Cystic swellings**: E.g., dermoid cysts.

Key Points to Note:

- Lack of **warmth and redness** is due to **suppressed immune response.**
- Always **rule out tuberculosis** in cases of chronic abscesses.

Q15. Causes of Persistent Sinus

- **Definition**:
  A **sinus** is an abnormal **blind tract** or channel opening onto a surface (skin or mucosa) and lined with **granulation tissue**. It remains **open and discharges pus or fluid**.

Causes

- **Infections**:
  - **Tuberculosis**: Chronic infection of bone (e.g., **tuberculous osteomyelitis**) or lymph nodes causing a sinus.
  - **Actinomycosis**: Deep-seated bacterial infection leading to chronic sinus with **sulfur granules**.
  - **Chronic osteomyelitis**: Persistent infection of the bone, commonly seen after **trauma or surgery**.
- **Foreign Body**:
  - Retained **foreign materials** like splinters, glass, or surgical sutures causing chronic irritation and infection.
- **Diseases of Bone**:
  - **Sequestrum** (dead bone fragment) in conditions like chronic osteomyelitis acts as a **nidus for infection**, leading to a persistent sinus.
- **Tuberculous Lymphadenitis**:

- Enlarged, **caseating lymph nodes** rupture to form a sinus, commonly in the **neck region**.
- **Inadequate Surgical Drainage**:
  - Improper **evacuation of abscesses** or deep infections can lead to sinus formation.
- **Pilonidal Sinus**:
  - Develops due to **ingrown hair** in the gluteal cleft, leading to chronic irritation and infection.
- **Congenital Abnormalities**:
  - Persistence of embryological remnants such as **thyroglossal duct cyst** or **branchial sinus**.
- **Trauma**:
  - **Penetrating injuries** or wounds that fail to heal properly may result in a sinus.
- **Malignancy**:
  - **Chronic sinus** associated with **squamous cell carcinoma** or other cancers in advanced stages.
- **Specific Conditions**:
  - **Crohn's disease**: Causes chronic sinus in the **perianal region**.
  - **Hydatid disease**: Ruptured hydatid cysts can lead to sinus formation.

Features of a Persistent Sinus

- Constant or intermittent **discharge** of pus or fluid.
- Presence of **granulation tissue** at the opening.
- **Underlying pathology** such as a **foreign body** or **infected tissue**.
- **Failure to heal** despite regular treatment.